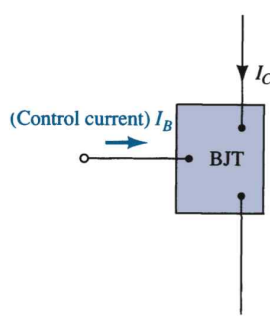
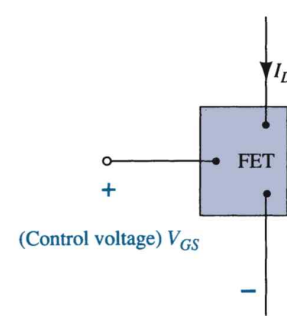


Chap. 6 Field-Effect Transistor (FET)

§6.1 Introduction

- Current-controlled / Voltage-controlled Amplifiers

BJT current-controlled device	FET voltage-controlled device
	

- FET and BJT

	BJT	FET
Similarities	can be used as Amplifiers can be used as Switching Devices	
Differences	1. Current Controlled Devices 2. Generally faster than FET	1. Voltage Contolled Devices 2. Higher Input Resistance 3. Less sensive to temprature variation than BJT 4. Better suited for ICs

- FET Types

1. JFET : Junction FET
2. MOSFET : Metal-Oxide-Semiconductor FET
3. D-MOSFET : Depletion-MOSFET
4. E-MOSFET : Enhancement-MOSFET

- suructure

1. PMOS : p-type MOS
2. NMOS : n-type MOS
3. CMOS : complementary MOS (PMOS+NMOS)

§6.2 Construction and Characteristics of JFETs

● JFET (Junction Field-Effect Transistor) Construction

There are two types of JFETs:

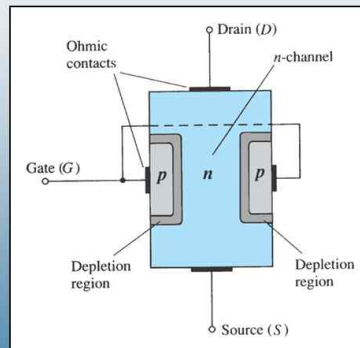
***n*-channel**
***p*-channel**

The *n*-channel is the more widely used of the two.

JFETs have three terminals:

The **Drain (D)** and **Source (S)** are connected to the *n*-channel

The **Gate (G)** is connected to the *p*-type material



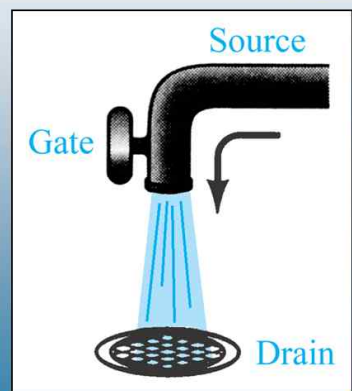
● JFET Operation: Basic Idea

JFET operation can be compared to that of a water spigot.

The **source** is the accumulation of electrons at the negative pole of the drain-source voltage.

The **drain** is the electron deficiency (or holes) at the positive pole of the applied voltage.

The **gate** controls the width of the *n*-channel and, therefore, the flow of charges from source to drain.

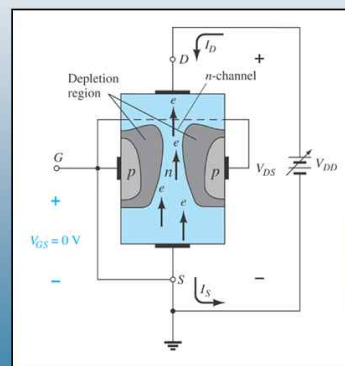


● JFET Operating (n-channel JFET)

($V_{GS} = 0\text{ V}$, V_{DS} Some Positive Value)

Three things happen when $V_{GS} = 0\text{ V}$ and V_{DS} increases from 0 V to a more positive voltage:

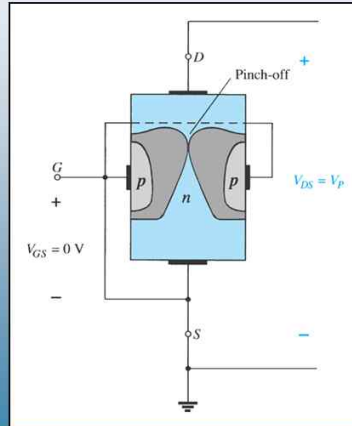
- The size of the depletion region between *p*-type gate and *n*-channel increases.
- Increasing the size of the depletion region decreases the width of the *n*-channel, which increases its resistance.
- Even though the *n*-channel resistance is increasing, the current from source to drain (I_D) through the *n*-channel is increasing because V_{DS} is increasing.



● Pinch-off

• If $V_{GS} = 0\text{ V}$ and V_{DS} continually increases to a more positive voltage, a point is reached where the depletion region gets so large that it **pinches off** the channel.

• This suggests that the current in channel (I_D) drops to 0 A, but it does not: As V_{DS} increases, so does I_D . However, once pinch off occurs, further increases in V_{DS} do not cause I_D to increase.

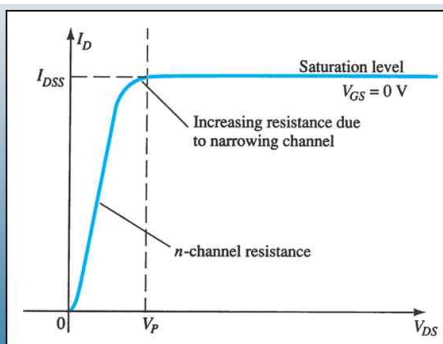


● I_D versus V_{DS} for $V_{GS} = 0\text{ V}$ (Saturation)

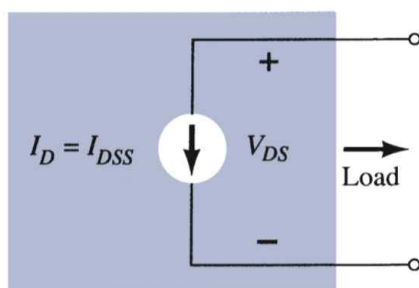
At the pinch-off point:

Any further increase in V_{DS} does not produce any increase in I_D . V_{DS} at pinch-off is denoted as V_P

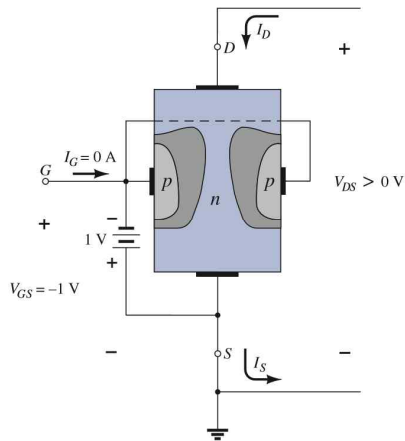
I_D is at saturation or maximum, and is referred to as I_{DSS}



※ I_{DSS} is the maximum drain current for a JFET and is defined by the conditions $V_{GS} = 0\text{ V}$ and $V_{DS} > V_P$. It is a kind of current sources.



● $V_{GS} < 0 \text{ V}$

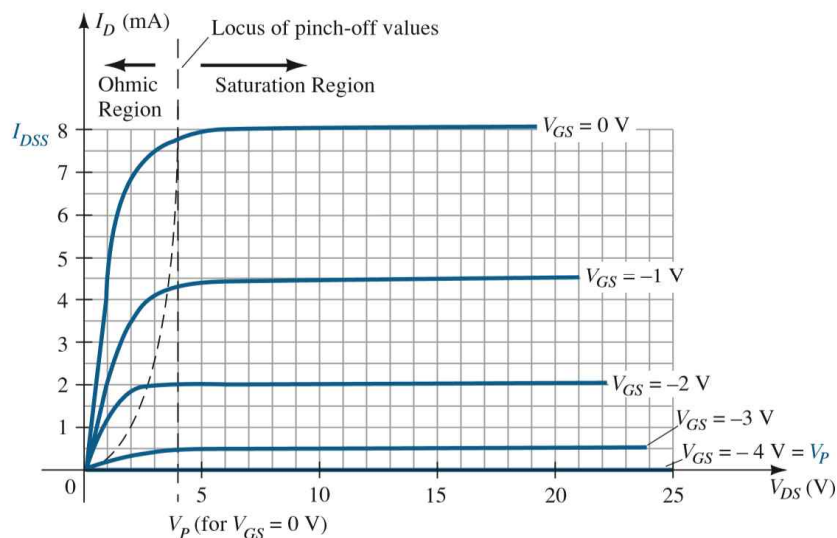


As V_{GS} becomes more negative, the depletion region increases.

The result of applying a negative bias to the gate is to reach the saturation level at a lower level of V_{DS} .

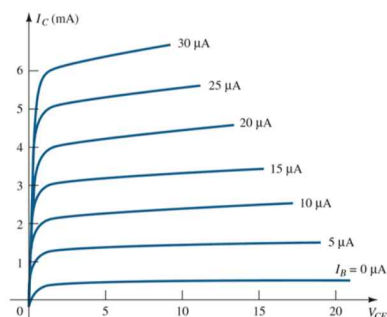
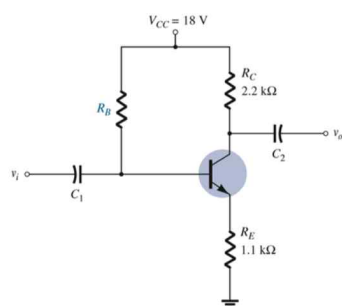
The saturation level for I_D has been reduced and in fact will continue to decrease as V_{GS} is made more and more negative.

● n-channel JFET characteristics (an example)



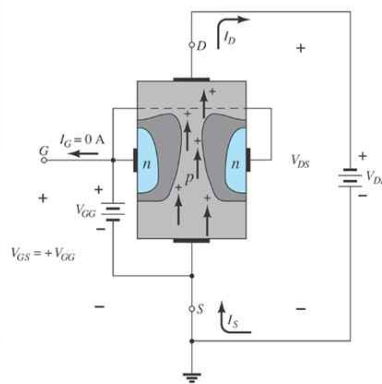
※ V_{GS} (input voltage) can control I_D (output current).

※ similar with the characteristics of BJT



- p-channel JFET

The *p*-channel JFET behaves the same as the *n*-channel JFET. The only differences are that the voltage polarities and current directions are reversed.



- JFET symbols

n-channel JFET	p-channel JFET

12/2

§6.3 Transfer Characteristics

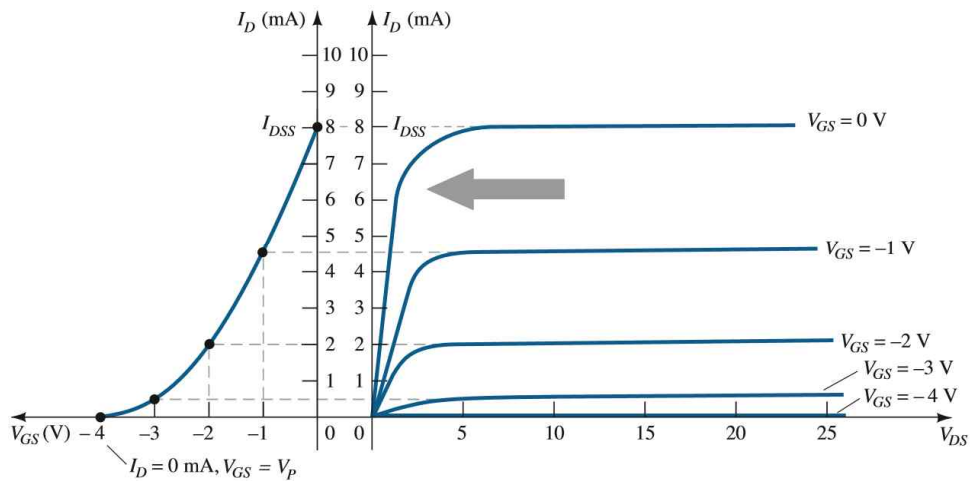
- JFET Transfer Characteristics

	JFET	BJT
control input	V_{GS}	I_B
output	I_D	I_C
relationship	$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$ (* Shockley's equation)	$I_C = \beta I_B$
remarks	I_{DSS} and V_P : constant	β : constant

※ JFET input-to-output transfer characteristics are not as simple as they are for a BJT.

● JFET Transfer Curve from the drain characteristics

(※ The graph shows the value of I_D for a given value of V_{GS} .)

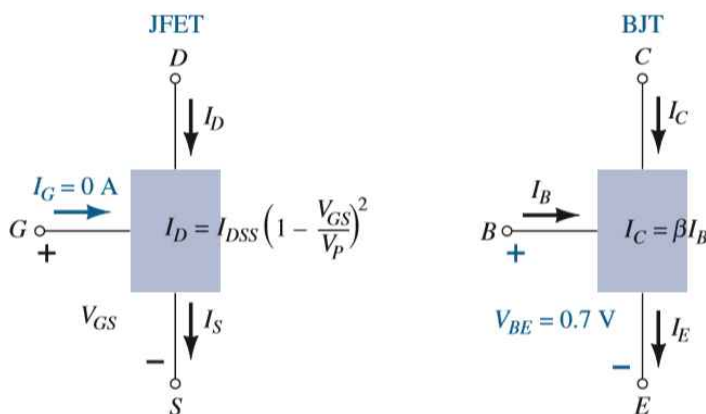


§6.4 Specification Sheet (JFET) Read

§6.5 Instrumentation skip

§6.6 Important Relationships

JFET		BJT
$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$	$\Leftrightarrow$	$I_C = \beta I_B$
$I_D = I_S$	$\Leftrightarrow$	$I_C \cong I_E$
$I_G \cong 0$ A	$\Leftrightarrow$	$V_{BE} \cong 0.7$ V



§6.7 Depletion-Type MOSFET

※ MOSFET : Metal-Oxide Semiconductor Field-Effect Transistor

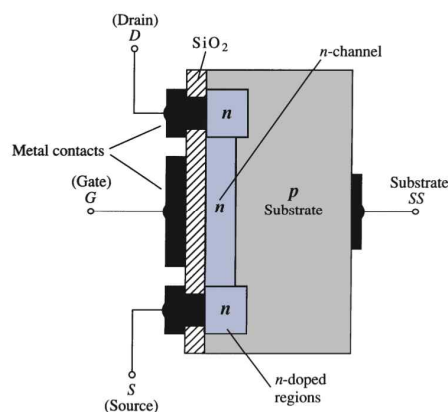
● MOSFET

- Characteristics are similar to JFETs.
- Two types of MOSFETs
 - Depletion-type MOSFET (D-type MOSFET)
 - Enhancement-type MOSFET (E-type MOSFET)

● Construction of n-channel depletion-type MOSFET

The **Drain (D)** and **Source (S)** connect to the n -type regions. These n -typed regions are connected via an n -channel. This n -channel is connected to the **Gate (G)** via a thin insulating layer of silicon dioxide (SiO_2).

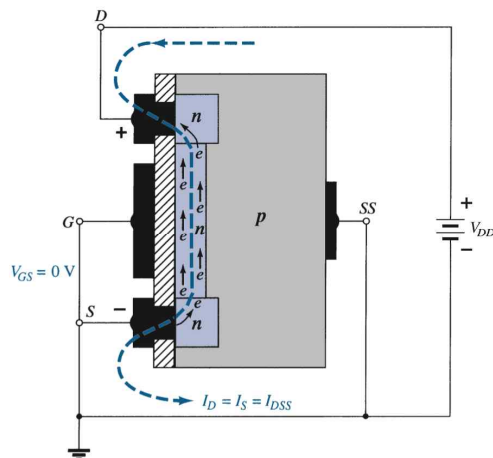
The n -type material lies on a p -type substrate that may have an additional terminal connection called the **Substrate (SS)**.



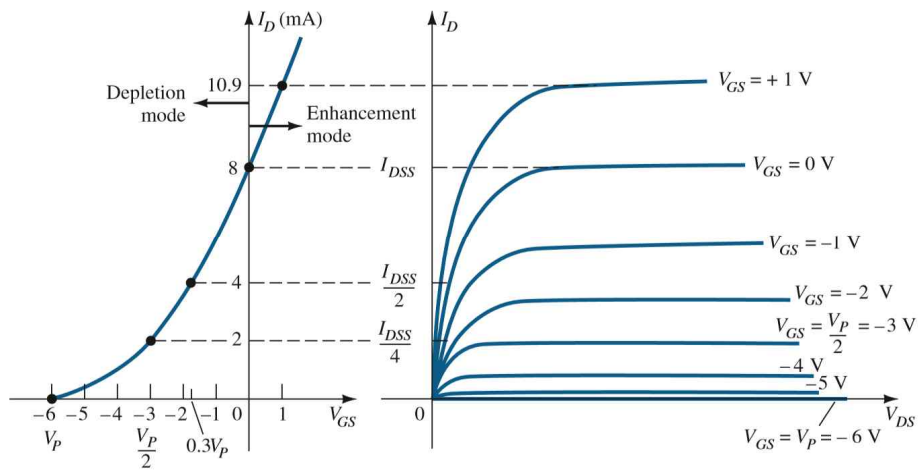
- ※ There is NO direct electrical connection between the gate terminal and the channel of a MOSFET.
- ※ It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.

● Basic Operation and Characteristics

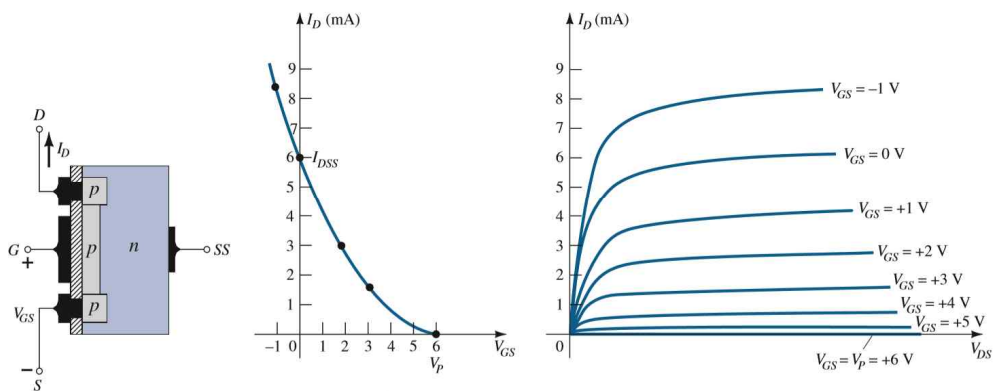
- n-channel depletion-type MOSFET with $V_{GS} = 0\text{ V}$



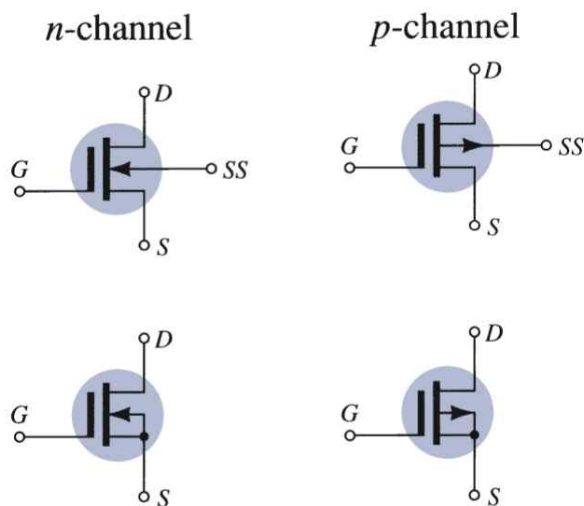
- Drain and Transfer characteristics
(* similar with the curves of JFET)



● p-channel Depletion-type MOSFET

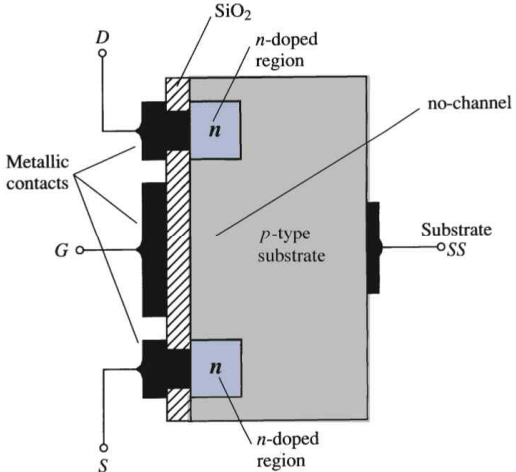
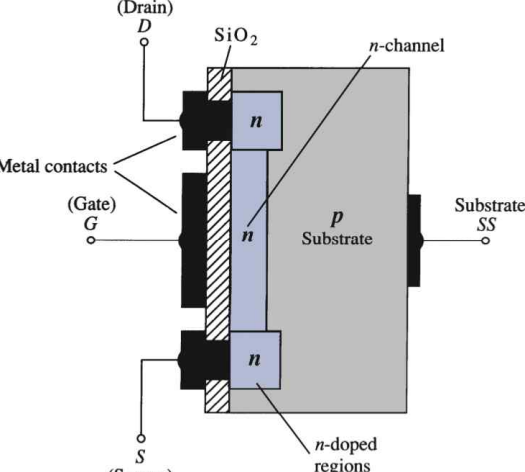


● Symbols of n-channel and p-channel depletion-type MOSFETs

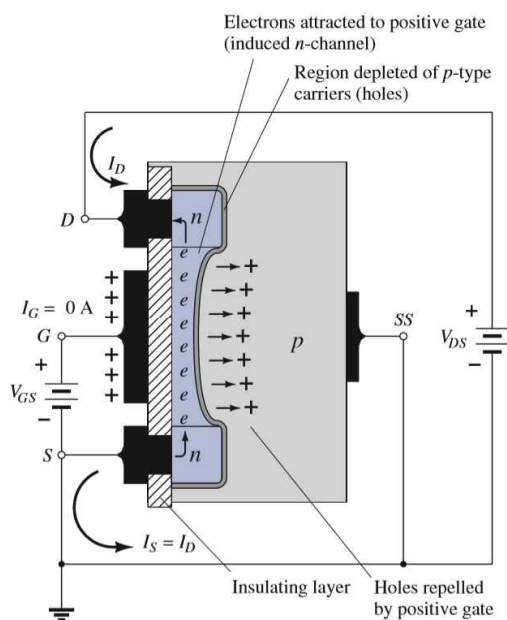


§6.8 Enhancement-Type MOSFET

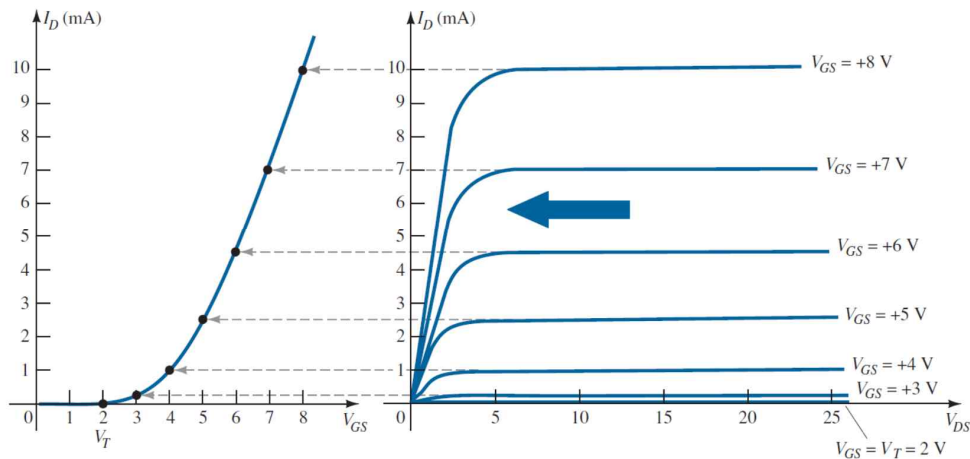
● Basic Construction (NO channel)

n-channel enhancement-type MOSFET (NO channel)	n-channel depletion-type MOSFET (n-channel)
	

● Basic Operation of n-channel enhancement-type MOSFET

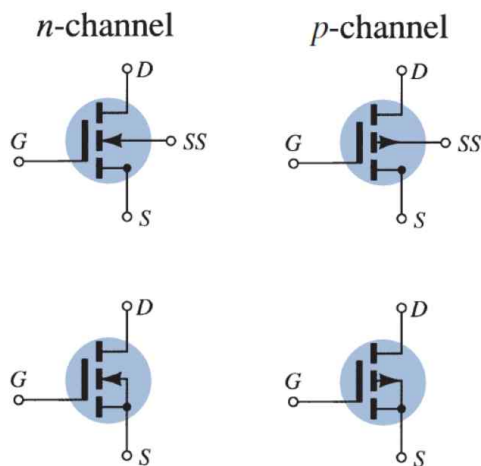


- Drain and Transfer characteristics
(n-channel enhancement-type MOSFET)



$$I_D = k (V_{GS} - V_T)^2 \quad V_T: \text{Threshold Voltage}$$

- Symbols of n-channel and p-channel enhancement-type MOSFETs



§6.9 MOSFET Handling

- MOSFETs are very sensitive to static electricity.

Because of the very thin SiO_2 layer between the external terminals and the layers of the device, any small electrical discharge can create an unwanted conduction.

- Protection from the damages
 - (a) Always transport in a static sensitive bag
 - (b) Always wear a static strap when handling MOSFETs

§6.10 VMOS and UMOS Power MOSFETs

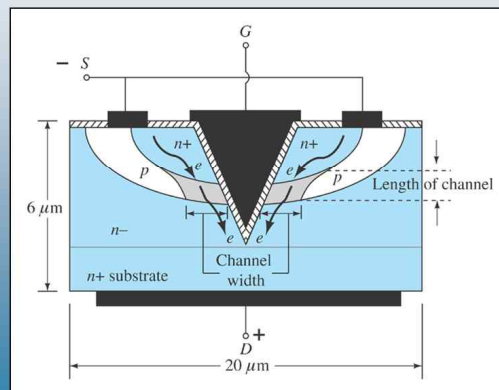
● VMOS MOSFET

VMOS (vertical MOSFET) is a component structure that provides greater surface area.

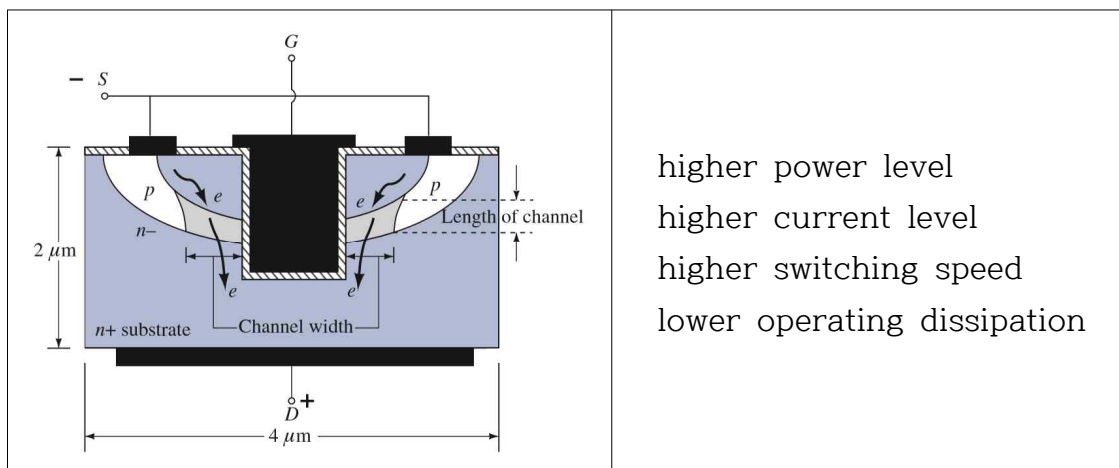
Advantages

VMOS devices handle higher currents by providing more surface area to dissipate the heat.

VMOS devices also have faster switching times.



● UMOS MOSFET

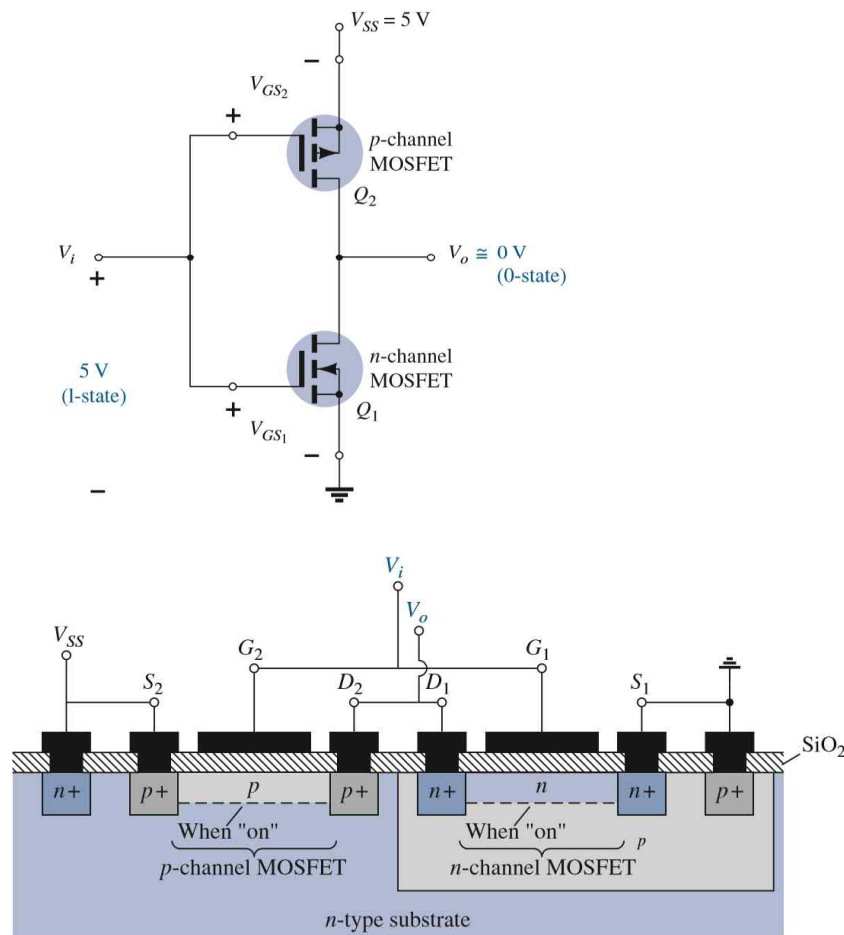


- ※ VMOS and UMOS devices typically have switching times less than one half that encountered for the typical BJT transistor.

§6.11 CMOS (Complementary MOSFET)

high input impedance
 high switching speed
 lower operating power levels
widely used for digital logic circuits

(Example) An Inverter for a digital logic circuit

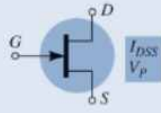
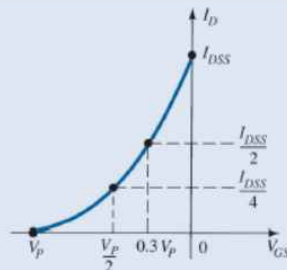
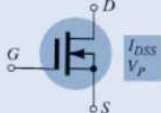
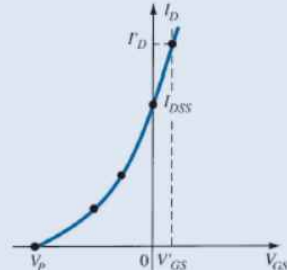
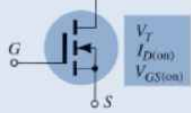
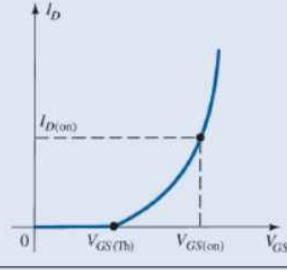


§6.12 MESFET skip

(※MESFET : METal-Semiconductor Field-Effect Transistor)

§6.13 Summary Table (Table 6.3) read

TABLE 3
Field Effect Transistors

Type	Symbol and Basic Relationships	Transfer Curve	Input Resistance and Capacitance
JFET (<i>n</i> -channel)	$I_G = 0 \text{ A}, I_D = I_S$  $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$		$R_i > 100 \text{ M}\Omega$ $C_i: (1 - 10) \text{ pF}$
MOSFET depletion type (<i>n</i> -channel)	$I_G = 0 \text{ A}, I_D = I_S$  $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$		$R_i > 10^{10} \Omega$ $C_i: (1 - 10) \text{ pF}$
MOSFET enhancement type (<i>n</i> -channel)	$I_G = 0 \text{ A}, I_D = I_S$  $I_D = k (V_{GS} - V_{GS(Th)})^2$ $k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$		$R_i > 10^{10} \Omega$ $C_i: (1 - 10) \text{ pF}$

§6.14 Summary read